Code No: 181AP **R22**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech I Year I Semester Examinations, September - 2023
PROGRAMMING FOR PROBLEM SOLVING
(Common to CSE, IT, CSIT, CE(SE), CSE(CS), CSE(AI&ML), CSE(DS), CSE(IOT), AI&DS, AI&ML, CSD)

Time: 3 Hours **Max. Marks: 60**

Note: This question paper contains two parts A and B.
i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.
- Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
- Part-B consists of **ten questions** (numbered from 2 to 11) **carrying 10 marks each**. From each unit, there are two questions and the student should answer one of them. Hence, the student should answer five questions from Part-B.

PART- A

(10 Marks)

1.a)	What is compiler?	[1]
b)	Give one example for syntax error.	[1]
c)	Define the enumeration data type in C.	[1]
d)	What is a two-dimensional array?	[1]
e)	How do you define a macro using the preprocessor?	[1]
f)	How do you append data to an existing file in C?	[1]
g)	What is recursion?	[1]
h)	What is the signature of a function?	[1]
i)	What is the key difference between linear search and binary search?	[1]
j)	What is the time complexity of selection sort?	[1]

PART - B

(50 Marks)

2.a)	Differentiate between object code and executable code in the context of compiling a program.	
b)	Discuss the different storage classes in C programming.	[5+5]
	OR	
3.a)	Discuss the different types of loops in C programming.	
b)	How can you pass command line arguments to a C program? Provide a step-by-step explanation.	[5+5]
4.a)	Differentiate between a null-terminated string and a fixed-length string.	
b)	Discuss the concept of a union and how it differs from a structure.	[5+5]
	OR	
5.	Discuss the significance of pointers in programming and provide examples of their practical usage in manipulating data.	[10]

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6.a) Compare and contrast text files and binary files in terms of their structure and usage.
b) Write a C program to read a text file line by line and display its contents on the console. [5+5]

OR

7.a) Describe the purpose of conditional compilation directives like #ifdef, #ifndef, and #endif in C.
b) Write a C program to read a text file and count the number of lines in it. [5+5]

8.a) Explain the process of freeing memory in C. What happens if memory is not freed after allocation?
b) Discuss the limitations of recursive functions. What factors should be considered before using recursion? [5+5]

OR

9. Describe the concept of passing parameters to functions in C with example programs. [10]

10.a) Write an algorithm for binary search and implement with an example.
b) Define Big O notation and explain its significance in analyzing algorithm efficiency. [5+5]

OR

11. Explain the bubble sort algorithm step by step with an example. [10]

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